

Selection of Harvestable Tea Shoots based on Height and Stiffness

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Abstract — Tea harvesting in Sri Lanka is extremely labor intensive and mechanization incorporated in the process is minimal. The main reason for this is the inability of existing harvesting devices to discriminate harvestable generations of tea shoots from the rest. To supersede manual harvesting, such devices should essentially harvest only tender tea shoots leaving the immature ones intact. In an ongoing research it was hypothesized that the height and stiffness of tea shoots are suitable parameters to distinguish harvestable tea shoots. A selection mechanism was then developed based the statistical information of the two identified parameters. Mathematical models were used to further improve its configuration, followed by experimental validation. Finally, field tests were conducted using a prototype to verify the accuracy of the proposed selection method. Test results showed promise and prompt further investigation.

Keywords — *discrimination criterion; height and stiffness of tea shoots; selective tea harvesting; selection mechanism*

I. INTRODUCTION

Sri Lanka is a leading tea producer in the world with a share of 17.3% of the world's tea exports in 2012 [1]. It is the main agricultural export product of the country [1]. Among the stages of tea production, tea harvesting is considered to be the most labor intensive phase. It accounts for 70% of the total workforce required for the production of made tea and contributes to more than 35% of total cost of production [2]. This is because tea harvesting in Sri Lanka is mainly carried out manually [2]. This manual operation is named as 'tea plucking' where laborers select and pluck the tea shoots ready for harvesting (i.e. harvestable shoots). Then the plucked shoots are collected in baskets or bags, which they carry on their backs [2].

There are six to seven distinguishable generations of shoots found over the plucking table on a typical tea bush in Sri Lanka [3], where a generation is considered as a family of shoots that are at the same level of growth. In order to produce high quality tea, the harvested shoots must be at the correct maturity level and a 'harvestable tea shoot' is a stem with a couple of tea leaves and a dormant bud that rise above the other shoots [2]. In a single harvesting round, one or two older generations are plucked, leaving the younger ones intact. This action is defined as 'selection'. The younger generations or immature tea shoots are called 'arimbus' and harvesting arimbus results in reduced quality of made tea and productivity of the tea bushes [3], [4]. Not plucking harvestable shoots also detrimentally affects the health of the bush in the long-term [4]. Although a skilled laborer plucks up to 20-30 kg of leaves per day [4], the process of selective plucking demands cumbersome labor.

Most tea fields within the country are located in the hill country on terrains that are primarily accessible by foot. It is due to the conducive weather conditions to grow high quality tea. As a result, tea plucking has been the choice for harvesting over the years [2]. Fortunately, Sri Lanka was blessed with the required workforce for manual tea harvesting at a cheap rate since the introduction of the South Indian laborers to work in the tea estates [5]. Availability of cheap labor also may have played a part in keeping the manual plucking process without moving towards mechanization.

In order for mechanization to be attractive, it needs to be both economical and effective than the manual method. The high cost and shortage of skilled labor in the present-day context have forced the industry to mechanize tea harvesting [4], [6]. Although several tea harvesting devices (harvesting shears and motorized machines) have been introduced to the market in the past, they lack the ability to perform selective harvesting [4], [6]. According to [6], 'conventional shears' used in Sri Lanka have achieved an output of 25-40 kg per man-day. However, due to harvesting of arimbus along with tender and/or mature shoots by the non-selective shears, the interval between plucking rounds had been extended to 10-14 days (where for manual harvesting, it is usually seven days) and their continuous usage has reduced the yield by 25-35 % in the long-run [6]. The 'TRI selective tea harvester (shear)' introduced by the Tea Research Institute (TRI) in Sri Lanka had later improved the harvesting efficiency by 50-100% resulting in an apparent labor saving of about 30% [6]. However, adoption rate of the device was found to be unsatisfactory. Nature of its manual operation, time taken for training and less output compared to motorized harvesters are the main reasons for its poor usage in the field [6]. In 'mechanical harvesting', the actual worker output is in the range of 20-60 kg per worker per day and the expected apparent labor saving is about 40%. However, motorized machines caused a significant crop loss of the magnitude of 30-50% as a result of bush debilitation and poor selectivity leading to reduction of quality of made tea [6].

The criteria for distinguishing different generations of tea shoots can generally be identified as color, size, shape, texture, height and stiffness of tea leaves/stems. However under mechanized selection, which is the scope of this research, the height and stiffness of the tea shoots are identified as probable parameters to be deployed. Therefore, the aim of this research was to develop a method to select harvestable shoots from the set of different generations of shoots present on local tea bushes using the shoot's height and stiffness. The objectives were to study the suitability of the two parameters for selection, to

develop a mechanism to select harvestable shoots and to evaluate the effectiveness of the proposed method.

II. METHODOLOGY

Statistical analyses using Student's t-test were performed by collecting height and stiffness data for both harvestable tea shoots and arimbus. In case of height data, five samples were gathered with 10 measurements from each. Heights of the randomly selected shoots of both harvestable shoots and arimbus were measured from the plucking table (imaginary surface of a tea bush which serves as a datum for plucking) to the top-most point of the shoots. The force to deflect was measured along the stems to determine the stiffness. In order to measure these parameters, test protocols and measuring tools were developed according to the procedure described in [7] and [8]. The investigation was limited to TRI 2026 tea clone cultivated in the intermediate-zone of the up-country region as it is one of the main cultivars [9] developed by the TRI of Sri Lanka. Afterwards, regression analysis was performed to model the shoot stiffness along the length of shoot for both harvestable shoots and arimbus. With the established knowledge, design alternatives were generated for a mechanical device to enable selection. Crude prototypes of these were constructed and then tested at tea fields [7]. A prototype was then developed using the most promising mechanism selected based on the experimental results, and further improvements to this were made using an analytical approach as discussed under subsections D and E of Section IV. After tuning the final prototype for smooth operation, it was fitted to a commercially available tea harvesting machine (model AM-110VB450 by Ochiai Tea Plantation & Harvesting Machines, Japan [10]) to conduct field tests and the percentage of harvested mature shoots and arimbus were calculated by counting them on the bushes before and after harvesting.

III. RESULTS AND DISCUSSION

A. Height measurements of tea shoots

Heights of different generations of tea shoots were assumed to be normally distributed and for harvestable shoots (mean: 153 mm, standard deviation: 48 mm) and arimbus (mean: 78 mm, standard deviation: 37 mm). The t-test revealed a significant difference between the heights of harvestable shoots and arimbus with a 95% confidence level ($p < 0.05$). The result can be further explained as follows. Assume that 95% of the harvestable shoots are to be selected. This can be achieved by selecting all shoots that are taller than 74 mm. At that characteristic height, 45% of arimbus will also get filtered out.

B. Stiffness measurements of tea shoots

To eliminate arimbus being harvested, a further filtering mechanism based on shoot stiffness was hypothesized. According to the analysis, the shoot stiffness can be modeled with two regression equations with a 95% confidence level ($p < 0.05$). The equations were developed using Minitab 15 by Minitab Inc..

$$F_h = 7.38 - 0.0432 h + 0.0938 d \quad (1)$$

$$F_a = 1.89 - 0.0245 h + 0.0862 d \quad (2)$$

Here, F_h and F_a are forces, measured in grams, required to deflect a harvestable shoot and an arimbu by a distance d mm at a height of h mm from the plucking table. Considering the relative importance, stiffness measurements were conducted for height ranges from 80-130 mm for harvestable shoots and from 40-60 mm for arimbus. Measurements were performed on randomly selected harvestable shoots and arimbus with mean heights. Difference between (1) and (2) gives rise to F_d , where,

$$F_d = 5.5 - 0.0187 h + 0.0076 d \quad (3)$$

Here, (3) indicates a notable difference between stiffness of harvestable shoots and arimbus. For example, at the characteristic height of 74 mm, to cause a deflection of 50 mm, the force to be exerted on a harvestable shoot is twice as much than that on an arimbu. For the selection procedure to be efficient, F_d must be large enough for the selection mechanism to identify the shoot type accurately. In other words, height at which a tactile filtering mechanism should operate must be as low as possible to improve its effectiveness.

Considering the above facts, a combined height and stiffness based selection criterion was proposed to further improve the ability to select harvestable shoots. Arimbus have lower stiffness compared to harvestable tea shoots. Thus, if a stiffness-based filtering method is employed, at the characteristic height, theoretically, it should select 95% of harvestable shoots while filtering all of the arimbus.

C. Design concepts

Subsequently, two different design concepts for filtering arimbus were developed. Details of both concepts and pilot tests carried out are explained below.

1) Concept-1: The notion was to use a flexible strip to differentiate the arimbus from harvestable shoot at the characteristic height level. Fig. 1, illustrates both the computer-aided design (CAD) model and the apparatus fabricated to perform the pilot tests. Since the stiffness of arimbus is substantially lesser than that of a harvestable shoot, a flexible strip can be engineered to resist the arimbus and prevent it from entering into the cutting zone. However, upon contact of a stiffer harvestable tea shoot, the strip flexes and opens the gate accordingly. Stiffness of the strip and surface finish are to be closely controlled to obtain the desired result. It is assumed that the impact forces are negligible and the slant angle of the strip can be changed to tune the degree of deflection as needed. Pilot test results were promising, but revealed several drawbacks. In particular, when a bunch of tea shoots comprising both harvestable shoots and arimbus approach the strip the apparent force increases. As a result, all shoots in that bunch end up in the cutting zone. In addition, arimbus which are too short tend to crawl under the strip and some of them find their way into the cutting zone. The quad-strip design shown in Fig. 2 was implemented with the intention of reducing the possibility of selecting arimbus as a result of tea shoot bunching-problem explained above.

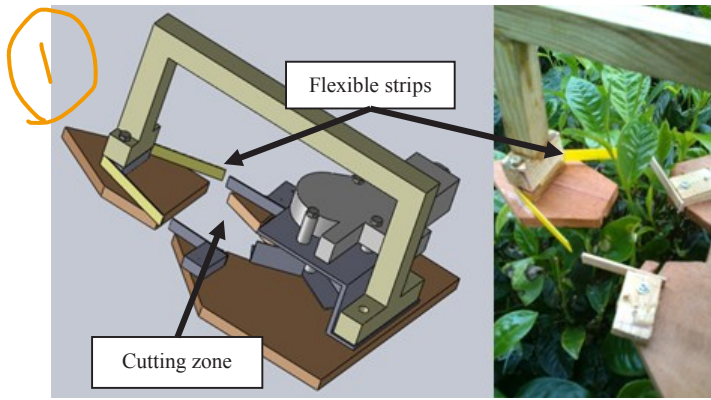


Fig. 1. Filtering arimbus using flexible strip(s).

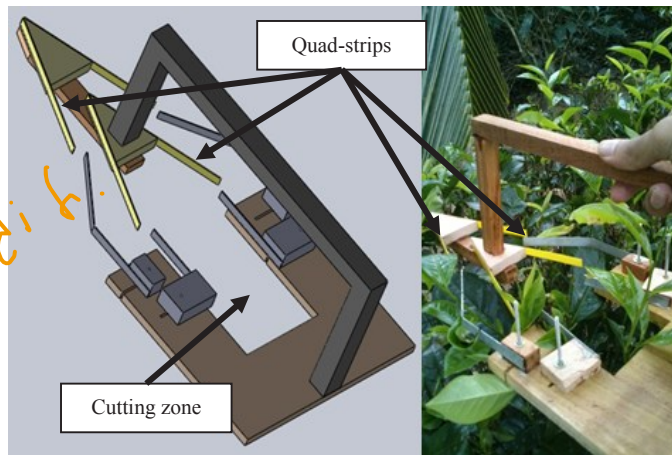


Fig. 2. Filtering arimbus using quad-strip design.

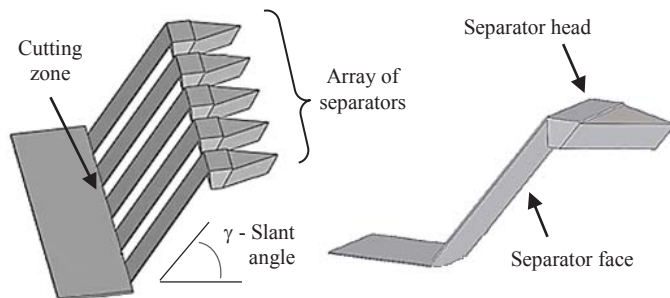
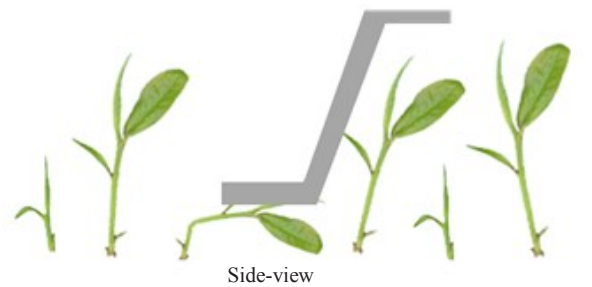


Fig. 3. Separator configuration of the comb-type filtering mechanism.

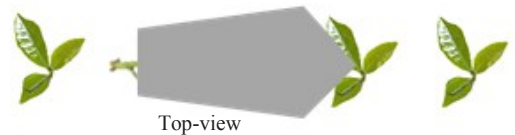
In the quad-strip design, each tea shoot that gets selected from the first-set of strips is tested again with the second-set of strips. At the second-set, most of the arimbus will get discarded and pushed away from the cutting zone. Pilot tests were conducted using custom made apparatus shown in Fig. 2, and the test results indicated no significant improvement over the previous design with dual-strip arrangement.

2) Concept-2: The notion was to use a set of separators with a comb-type configuration to filter arimbus from the rest of the tea shoots. The basic structure of the prototype is shown in Fig. 3. Unlike in concept-1, this mechanism can operate over the plucking table directly, and therefore, it is capable of selecting

all harvestable shoots with higher stiffness. Here, the array of separators and separator heads are the main components, which help to perform stiffness-based selection. The shearing mechanism is located at the space between the separators, which is labeled as the cutting zone. During operation, the separator heads are pushed parallel to the plucking table at 74 mm height level with the cutting mechanism kept aligned with the plucking table. As depicted in Fig. 4, short or shoots with low stiffness such as arimbus shall bump on to the separator face or separator head and sneak underneath the cutting mechanism. Since taller arimbus are more compliant, it is assumed that upon the frictional forces, they may laterally bend and crawl under. However, when the taller or stiffer harvestable shoots come into contact with the edges of the separator head, they will slide along the separator into the cutting zone as illustrated in Fig. 5. The width of the separator must be controlled to minimize the degree of lateral bent of harvestable shoot and thereby avoid any considerable bending that might cause it to crawl under the separator and vice versa for the arimbus.



Side-view

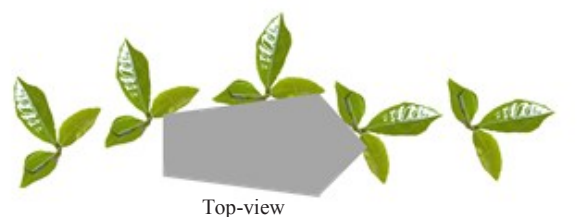


Top-view

Fig. 4. Function of separator(s) on arimbus.



Side-view



Top-view

Fig. 5. Function of separator(s) on harvestable shoots.

D. Design of the prototype

The key design parameters considered for the prototype were distance between separator heads (w), height of separator head (h), edge length of separator head (L), included angle between edges of separator head (α), slant angle of separator face (γ) and speed of travel (U) of the harvesting machine over the plucking table (see Fig. 3 and Fig. 6). The separators were planned to be equally spaced and total width of the gap open to the cutting mechanism was approximately half the width of the prototype. Assuming both harvestable shoots and arimbus are homogeneously distributed over the bush, proposed mechanism should ideally select all harvestable shoots while filtering 50% of arimbus. This is because, the arimbus that approach the gap between separators are allowed to venture into the cutting zone without being filtered. The suitable angle between edges of the separator head was identified by mathematically modeling the system and assuming or experimentally determining the variables that are perceived as less significant.

E. Mathematical modelling

The mathematical formulation to determine the effective angle of the separator head is explained below. Suppose the separator head is held at the characteristic height ($h = 74$ mm) and pushed over the plucking table at a speed of U m/s. As shown in Fig. 7, the speed of the separator edge can be divided into two component speeds.

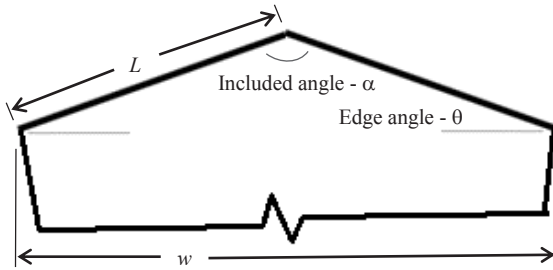


Fig. 6. Top-view of separator head.

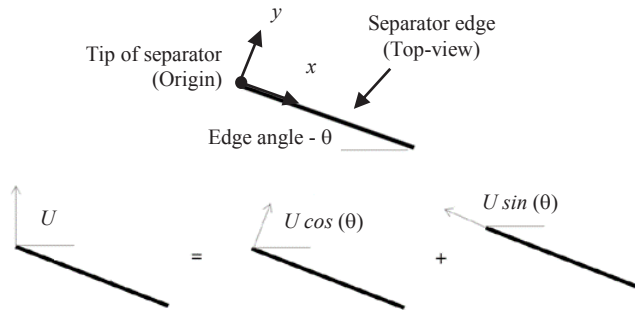


Fig. 7. Principle of superposition for velocities.

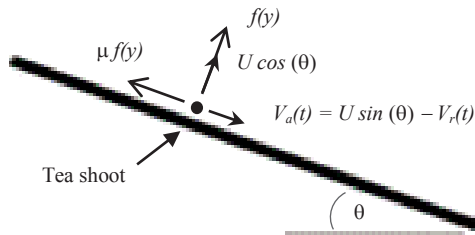


Fig. 8. Forces and velocities observed by the shoot relative to the edge.

For simplicity, effects on the system due to motion in both X-direction and Y-direction were assumed linear. Hence, the principle of superposition can be applied to this system. As a result, the movement of the tea shoot due to the initial speed U is equal to the sum of the two effects due to the component velocities. Consider a harvestable tea shoot that has engaged with the separator tip and is sliding along the separator edge. As the edge moves in the Y-direction at a speed of $U \cos(\theta)$ the tea shoots will also be pushed simultaneously. Therefore, at time t , $x(t)$ and $y(t)$ are the x and y coordinates of the contact point of the tea shoot with respect to the origin. The normal force between the shoot and the edge is $f(y)$, which is calculated from the regression equation for deflection-force of harvestable shoots. Consequently at time t ,

$$y(t) = U t \cos(\theta) \quad (4)$$

$$f(y) = 0.041 + 0.919 y(t) \quad (5)$$

Here, $f(y)$ and $y(t)$ are measured in Newton and meter units respectively. As depicted in Fig. 8, the absolute velocity of the tea shoot along the X-direction can be expressed as,

$$V_a(t) = U \sin(\theta) - V_r(t) \quad (6)$$

Where, $V_r(t)$ is the relative velocity of tea shoot with respect to the separator edge caused by the frictional forces during sliding. Considering the aforementioned relative frame, Newton's second law can be applied to the tea shoot to determine $V_r(t)$ or $\ddot{x}(t)$ along X-direction.

$$m \ddot{x}(t) = \mu f(y) \quad (7)$$

Equations (4), (5) and (7) can then be used to derive an expression for $\ddot{x}(t)$.

$$\ddot{x}(t) = \mu (k / m) + U \cos(\theta) (c / m) t \quad (8)$$

Here, $\mu = 0.47$ is the coefficient of friction and $m = 0.003$ kg is the mean mass of a harvestable tea shoot. Each parameter had been determined experimentally using a custom made test apparatus and a scale. The statistical significance was also verified for the number of samples selected. The travel speed U of the separator edge was assumed to be 0.5 m/s. The coefficients $k = 0.041$ and $c = 0.919$ are retrieved from (5). The differential equation is then solved using initial conditions $x(0)=0$ and $\dot{x}(0)=0$. Finally, the expression for $x(t)$ can be developed as follows:

$$x(t) = \mu (k / 2m) t^2 + U \cos(\theta) (c / 6m) t^3 \quad (9)$$

Let L be the length of the edge of the separator head. Also suppose that the shoot traverses the edge in the time period from $t = 0$ to T . Then,

$$L = \int_{t=0}^{t=T} (U \sin(\theta) - V_r(t)) dt \quad (10)$$

Solving the integral equation leads to the following expression.

$$L = U \sin(\theta) T - x(T) \quad (11)$$

It is assumed that the tea shoot will deflect about a reference point located 10 mm beneath the plucking table. The maximum allowable deflection angle of harvestable tea shoot is assumed to be 15° . As a result, $y(T) = 22.5$ mm considering tea shoot deflection is linear over its height. Now, using (4) expression for T can be derived as follows:

$$T = 0.045 / \cos(\theta) \quad (12)$$

Subsequently, to cause an angular deflection of 15° the tea shoot should traverse the entire length of the separator head over its edge. Therefore,

$$L = 0.0225 / \cos(\theta) \quad (13)$$

By applying (9), (12) and (13) in (11) and substituting appropriate values for variables, and expression for edge angle (θ) is developed.

$$22.5 / \cos(\theta) - 45 \tan(\theta) + 8.84 / \cos^2(\theta) = 0 \quad (14)$$

Using 'fsolve' function in Matlab R2015a by Matlab Inc., (14) was solved numerically to find θ and its value was 56° . As a result, included angle between the edges of the separator head is $\alpha = 68^\circ$. Now, using (13) L is found to be 40 mm in length. Thus, the spacing between any two separator heads was set equal to the width of a separator; $w = 45$ mm. The slant angle of separator face was set at $\gamma = 30^\circ$.

F. Performance indicators

Validity of the mathematical formulation was investigated at the tea field using basic prototypes of the proposed mechanism as depicted in Fig. 9. The key performance indicator used for the investigation are based on the probability of correctly or incorrectly selecting harvestable shoot or arimbu respectively. Definitions for probabilities of selecting a harvestable shoot (P_h) and of selecting an arimbu (P_a) are expressed as follows:

$$P_h = h / H \quad (15)$$

$$P_a = a / A \quad (16)$$

Where,

H – Number of harvestable shoots in area before selection

h – Number of harvestable shoots selected to be harvested

A – Number of arimbu in area before selection

a – Number of arimbu selected (erroneously) to be harvested

G. Field tests

The experiment was initiated by counting the total number of harvestable shoots and arimbu on the bushes in a given area. Then the separator heads of the basic prototype were pushed parallel to the plucking table at an approximate speed of 0.5 m/s and at a 74 mm height. Tea shoots that reached the cutting zone (vacant space between separators) were carefully counted while noting the types of shoots getting selected. The analysis of the data gathered revealed promising results for the optimized variables derived from the mathematical model. For example, probabilities of selecting harvestable shoots and filtering arimbu were 100% and 56% respectively.

A sensitivity analysis on the edge angle (θ) was also carried out to critically evaluate the model results obtained from the mathematical formulation. Field tests were conducted using several basic prototypes prepared with varying edge angles in steps of 5° from 35° to 65° . Fig. 10, illustrates the variation of key performance indicators with edge angle. Notably when $\theta = 56^\circ$, the prototype outperformed the rest when it came to selecting harvestable shoots. However, percentage of arimbu erroneously selected was relatively high. Considering the experimental results, a better compromise could be reached by setting $\theta = 65^\circ$, as the percentage of arimbu saved was as high as 65% and percentage of harvestable shoots getting selected was 99%.

Note that all results were from controlled experiments. Therefore as the next step, a full-scale prototype with an array of separator heads were developed as illustrated in Fig. 3. The complete prototype or the gauging device was customized to fit on to a commercially available tea plucking machine known as Ochai AM-110VB450 [10].



Fig. 9. Performance investigation of the basic prototype at the tea field.

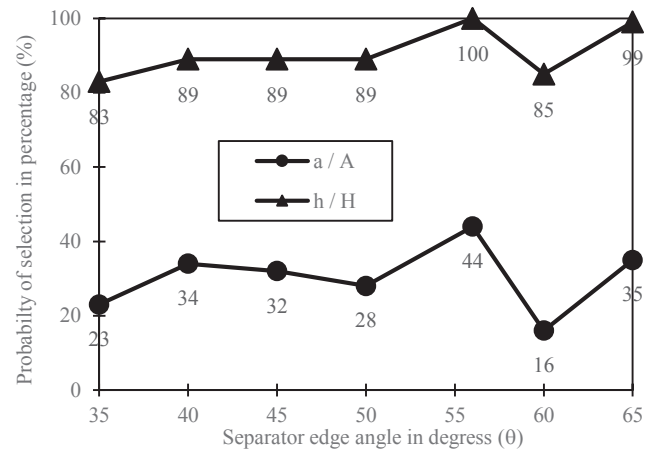


Fig. 10. Performance variation with edge angle.

The attachment was fabricated using aluminum alloy sheets, as the material is both weather resistant and lightweight. The overall length of clipper blades on the machine was 450 mm, therefore a set of five separators, each 65 mm width were mounted on the device equally apart, where first and the last separator heads were aligned with the two ends of the blade. The spaces between the separators were also increased to avoid cluttering of the tea shoots observed during experiments with the basic prototypes.

Fig. 11, shows the complete prototype having five-separator layout arrangement prepared for field testing. The device was attached on the frame of the shear-type tea plucking machine. Weight of the attachment was approximately 10% (1 kg) of that of the weight of the machine. The prototype test results showed that the mechanism is capable of selecting 52% of the harvestable shoots while filtering out 48% of the arimbus on average. Results were obtained by counting the types of shoots on the bushes before and after harvesting. However, these test results are only valid for TRI 2026 tea clone cultivated in intermediate-zone of the up-country region.

Notable advantage of the proposed gauging mechanism is that it can be readily customized and be fitted on to different types of non-selective tea harvesting machines. It is proposed to fabricate the final device using a polymer material such as thermoplastic (i.e., ABS) or carbon fiber reinforced composites. In particular, to use additive manufacturing (i.e., 3D printing) or rapid prototyping methods to develop the next level prototype having provisions for adjustments. Thereby, the structure will become lightweight and compact. The strength of the attachment should be sufficient enough to withstand the applicable forces during transportation and handling of the machine at tea fields. The comb-type mechanism also showed a few weaknesses. At times when it encountered a bunch of tea shoots, all shoots entered into the cutting zone as a result of increased apparent stiffness. There were instances where tea shoot as bunches got stuck (i.e., cluttering) and bent under the separators. These observations suggests the need for further improvement of the filtering mechanism.

The developed regression models are only applicable to the conditions of the tea plantation where the studies were carried out and it is understood that the characteristics of tea bushes vary according to the tea clone, weather conditions and other environmental factors. Therefore, it is essential that any mechanism, which is designed to select harvestable shoots, utilize an appropriate adjusting mechanism to change the cut-off values of selection with regard to both height and stiffness. This adjustment is essential in the Sri Lankan context because tea shoots are extremely sensitive to rain [3] and the same mechanism can be used during the rainy season as well as in the dry season if cut-off values can be changed. The experiments for evaluation of the attachment were carried out in the estates during dry environmental conditions. However, the regions in which tea is grown receive regular rainfall [2]. Therefore, the attachment also needs to be checked for its performance during wet conditions.



Fig. 11. Completed prototype fitted on to a tea plucking machine.

IV. CONCLUSION

Both height and stiffness are two potential parameters that can be used to select harvestable tea shoots. The developed device as an attachment to an existing harvesting machine showed promise with a capability of 52% selection of harvestable shoots and filtering out 48% of arimbus. However, extensive field tests should be performed to generalize the findings and to refine the proposed method.

Acknowledgments

Authors are thankful to Dr. M. A. Wijeratne and his staff at TRI, Ratnapura, Sri Lanka for sharing their knowledge and experience on existing tea harvesting machines. We are grateful to the governing authority for providing permission to perform experiments at tea plucking fields owned by the TRI and providing staff to assist the experimental work.

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